Earth and Planets · Teacher and Cover Plan

Learning goal: I can describe the Sun, Earth, and Moon as spheres and explain what orbits what in our solar system.

Scheme Outcome

TRACK Science scheme of work: Earth and Space

Describe the Sun, Earth, and Moon as approximately spherical bodies and describe the movement of the Earth and other planets relative to the Sun in the solar system.

Success Criteria

I can state that the Sun, Earth, and Moon are approximately spherical bodies.

I can order the eight planets from the Sun and distinguish rocky planets from gas/ice giants.

I can explain that planets orbit the Sun while the Moon orbits the Earth.

Model Sequence (I do)

First, the Sun is a star at the centre of the solar system, not a planet.

Second, the four inner planets (Mercury, Venus, Earth, Mars) are smaller and have rocky surfaces.

Third, the four outer planets (Jupiter, Saturn, Uranus, Neptune) are much larger gas and ice giants.

How do we know Earth is a sphere and not flat? Ships disappearing hull-first over the horizon provided

early evidence.

During a lunar eclipse, Earth casts a curved, circular shadow on the Moon from every angle.

Today, satellite photographs from space directly show the Sun, Earth, and Moon as spherical bodies.

We Do & Hinge Check

Which statement correctly compares the movement of the Earth and the Moon?

Answer: A. Earth is a planet orbiting the Sun; the Moon is a natural satellite orbiting the Earth.

Safe Practical & Card Setup

Order the eight planet cards from the Sun and build the Sun-Earth-Moon orbit diagram.

Kit: Printed Solar System Planet Cards, Sun-Earth-Moon Orbit Mat, and 3D sphere models (ball, marble, bead).

Never look directly at the real Sun in the sky. Keep small marbles/beads on the desk tray.

You Do Routes (Supported, Independent, Aspire)

Supported: Label the Sun, Earth, and Moon on the orbit diagram and circle the four rocky planets. (Help: Centre star = Sun. Planet = Earth. Satellite = Moon. First four = Mercury, Venus, Earth, Mars.)

Independent: Write the eight planets in order from the Sun, draw a labelled diagram showing Earth orbiting the Sun and the Moon orbiting Earth, and give one piece of evidence that Earth is a sphere. (Help:

Order: M__, V__, E__, M__, J__, S__, U__, N__. The __ orbits the Sun, while the __ orbits the Earth.)

Aspire: Explain two pieces of evidence that Earth is spherical and explain why poster diagrams of the solar system are not drawn to scale. (Help: Evidence includes __ and __. Posters are not to scale because the Sun is over __ times wider than Earth and the distances are vast.)